

Critical Analysis of AI Applications in Communication

The influence of persona and conversational task on social interactions with an LLM-controlled embodied agent

Presenters: Dice

➔ Research Background & Objective

Core Objective

Exploring how an LLM-controlled embodied agent's persona and dialogue tasks influence user behavior and experience in VR.

Theoretical Framework

CASA (Computers as Social Actors) – users apply human social rules to computers.



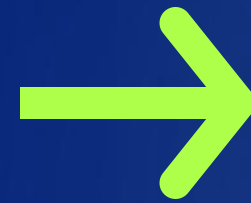


Experimental Design

46 participants engaged in face-to-face VR interactions with a male LLM agent.

Independent Variables

- **Persona**
(Extraverted vs. Introverted)
- **Tasks**
(Small Talk, Knowledge, Convincing)



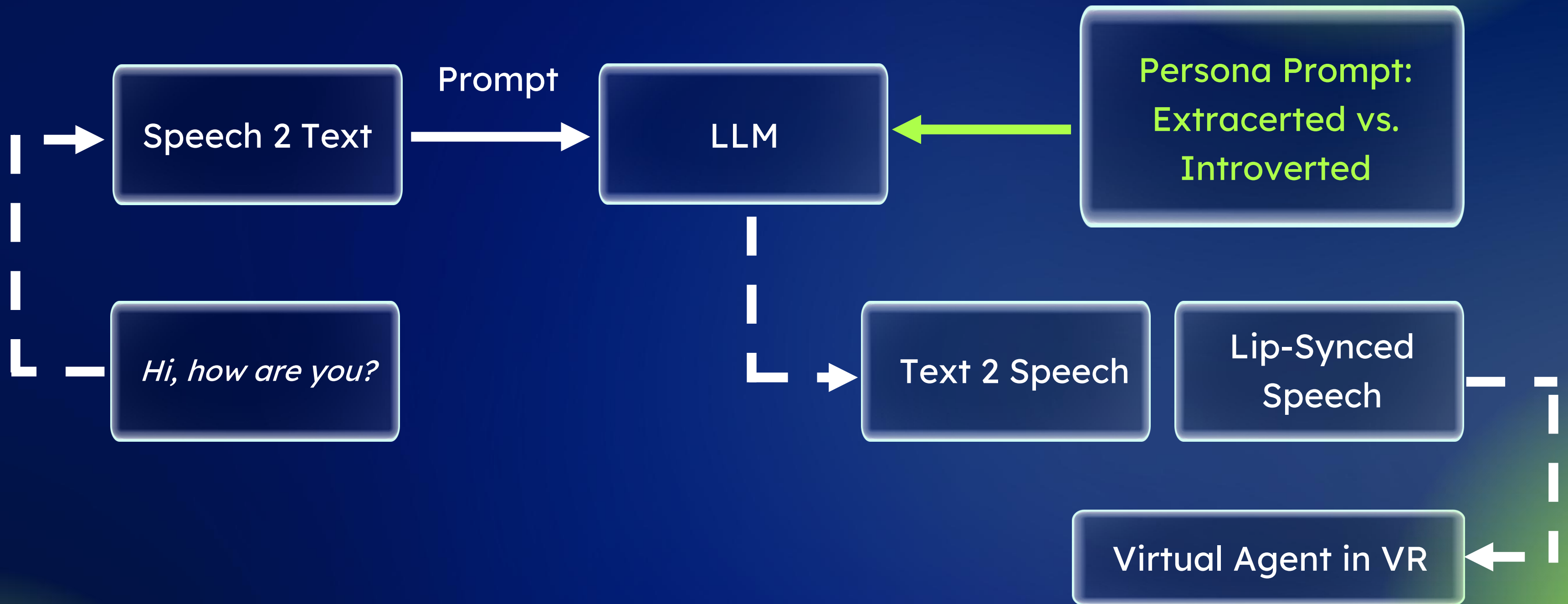
Dependent Variables

- Social Evaluation
- Emotional Experience
- VR Interaction Experience
- Behavioral Engagement
- Help-Seeking in tasks

Core Conclusion: Extraverted > Introverted

(Exception: The persona factor was insignificant when users needed to ask for help during the Knowledge Test)

➔ The AI Technical Pipeline



➔ The AI Technical Pipeline

Speech Input

Real-time voice capture mapped through Speech-to-Text modules.

LLM

A local LLM (SauerkrautLM) acts as the agent's brain for response generation.

Parameter Tuning

Researchers dynamically shaped the AI's personality by adjusting specific LLM parameters (e.g., Temperature and Sentence length constraints).

VR Avatar

Text-to-Speech output synced with Nvidia Audio2Face for real-time, face-to-face lip-syncing in VR.

Strength 1 - AI Methodology Innovation

Core Strengths

- Systematic prompt engineering for LLMs
- Explicit personality description prompts
- Adjusted specific hyperparameters (Temperature & Max sentence length)
- Highly systematized technical architecture

Positive Impact

- Successfully distinguished between extroverted and introverted agents
- Created a controllable, dynamic AI interaction pipeline

➔ Strength 1 - AI Methodology Innovation



Contribution: Validated via pre-tests and manipulation checks, offering a reliable novel method for human-AI interaction.

Strength 2 - Mixed-Methods Approach

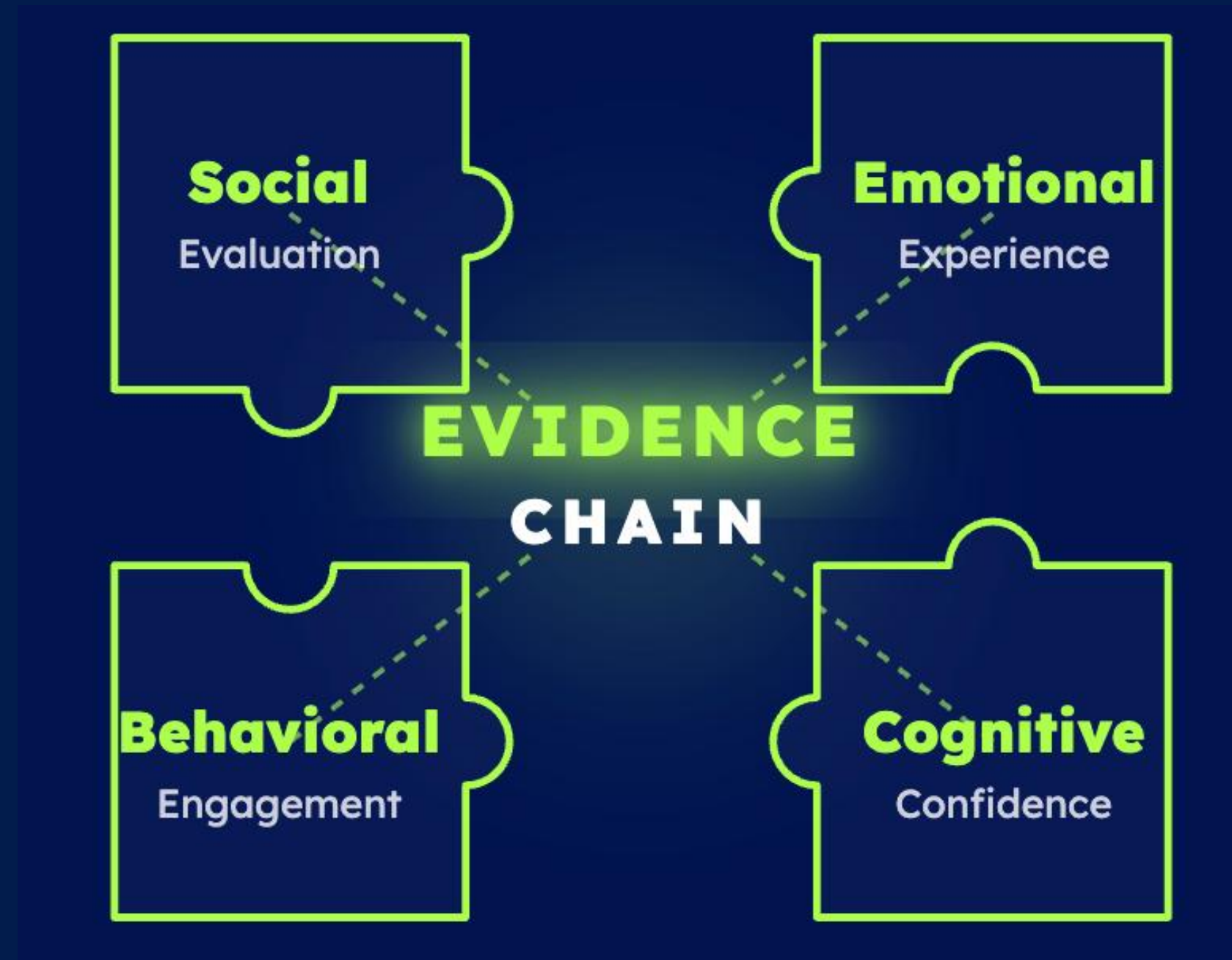
Core Strengths

- Comprehensive mixed-methods approach
- Integrated both self-reported and behavioral data
- Multi-level data collection

Positive Impact

- Enabled dynamic interactive engagement analysis
- Captured multi-layered user responses (objective actions vs. subjective feelings)

➔ Strength 2 - Mixed-Methods Approach



Contribution: Forged a robust evidence chain seamlessly linking social, emotional, behavioral, and cognitive dimensions.

Strength 3 - CASA Framework Validation

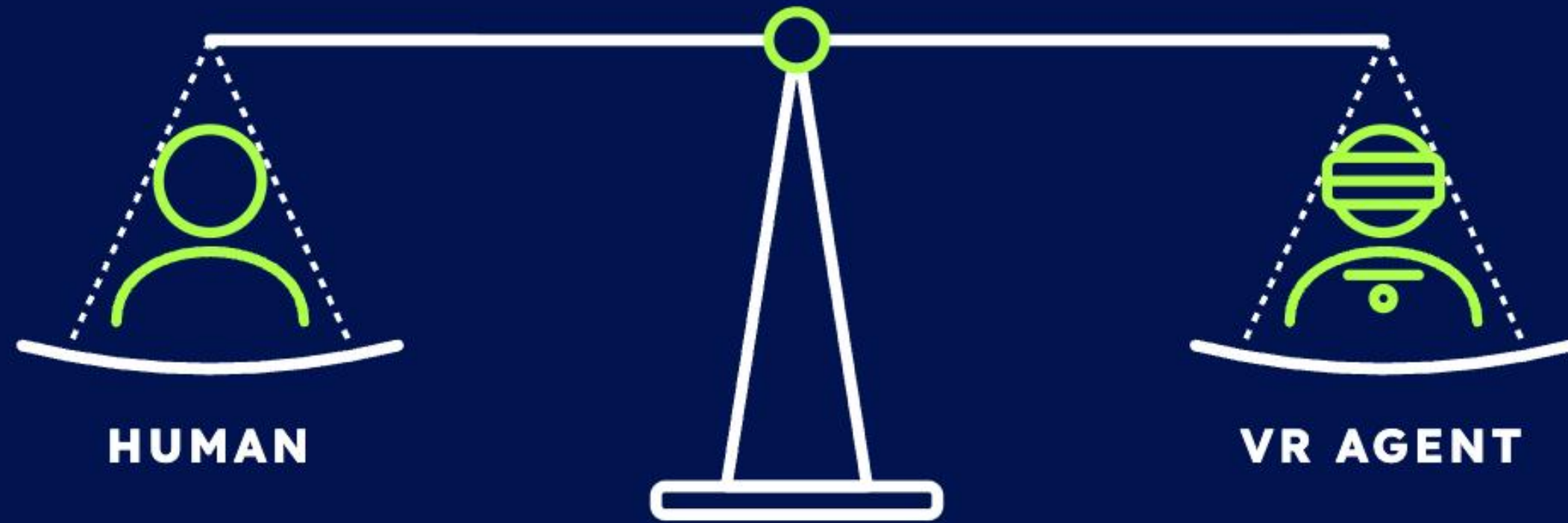
Core Strengths

- High-fidelity, immersive VR environment
- Three conversational tasks reflecting real-world social scenarios
- Robust validation of the CASA (Computers Are Social Actors) framework

Positive Impact

- Confirmed users apply human social principles to virtual agents.
- Task-dependent nuances revealed (personality effect strongest in persuasion)

➔ Strength 3 - CASA Framework Validation



SAME SOCIAL PRINCIPLES

Contribution: Proved the moderating role of task contexts, successfully bridging traditional HCI theory and modern LLM avatars.

➔ Weakness 1 - AI Technique & Embodiment

Core Limitations

- Small 7B-parameter local model - SauerkrautLLM
- Deliberately excluded conversation history
- Manually set hyperparameters (Temperature & Max sentences)
- Deficient physical embodiment: Only eye gaze & lip sync

Negative Impact

- Prevents natural multi-turn dialogue
- Lacks multimodal social cues
- Risks triggering the "Uncanny Valley" effect

➔ Weakness 1 - AI Technique & Embodiment



NO MEMORY & NO GESTURE

Consequence: Ultimately tested a "stiff talking head" rather than presenting a truly context-aware, realistic social agent.



Weakness 2 - Narrow Sample

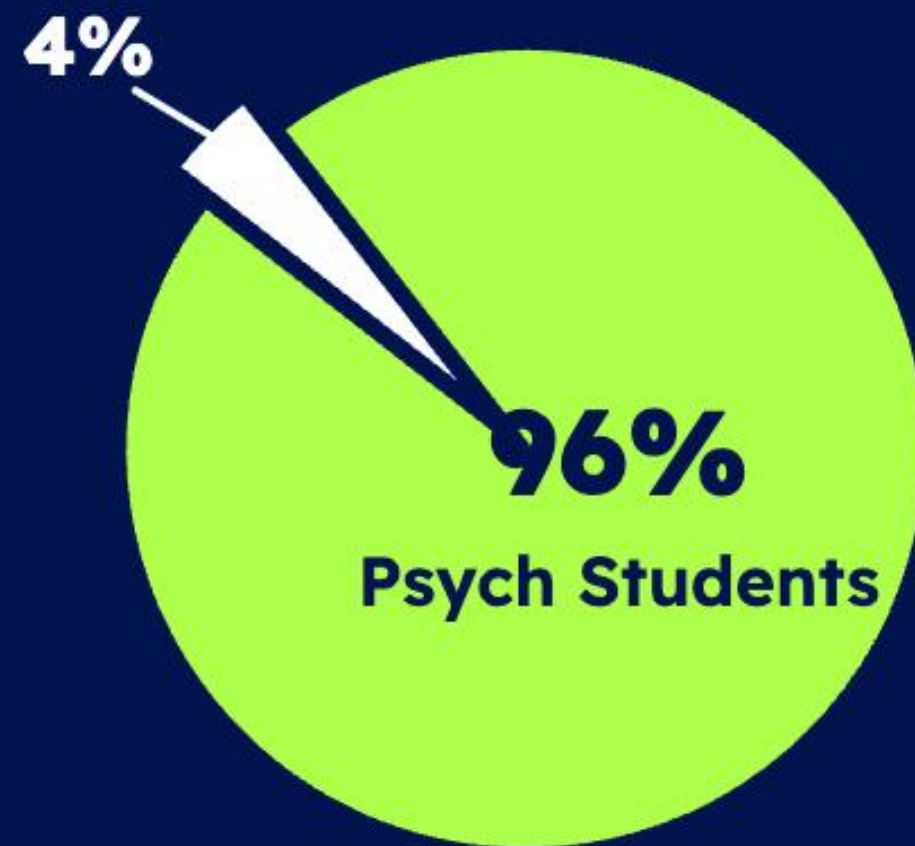
Core Limitations

- Highly skewed, strictly "WEIRD" demographic bias
- Only 46 participants (Average age: 21.24)
- 78% female & 96% psychology students
- 100% Western European

Negative Impact

- Ignores major extraversion differences between Western (Individualistic) and Eastern (Collectivistic) cultures
- Ignores sub-dimensions heavily influenced by age and gender

➔ Weakness 2 - Narrow Sample



W.E.I.R.D.

(Western, Educated, Industrialized,
Rich, Democratic)

Consequence: The findings are not universally valid, remaining applicable only to young, Western, female psychology students.

Weakness 3 - Baselines & Task Order

Core Limitations

- Absence of baseline comparisons (No broader frontier models or predefined approximations)
- Rigid task sequence performed in a fixed order: Chatting → Knowledge Test → Persuasion

Negative Impact

- Ignores major extraversion differences between Western (Individualistic) and Eastern (Collectivistic) cultures
- Ignores sub-dimensions heavily influenced by age and gender

➔ Weakness 3 - Baselines & Task Order



THREAT TO INTERNAL VALIDITY

Consequence: With zero counterbalancing, it is impossible to separate true task effects from order bias, seriously compromising internal validity.

➔ Improvement 1 - Contextual Memory Integration

Solution

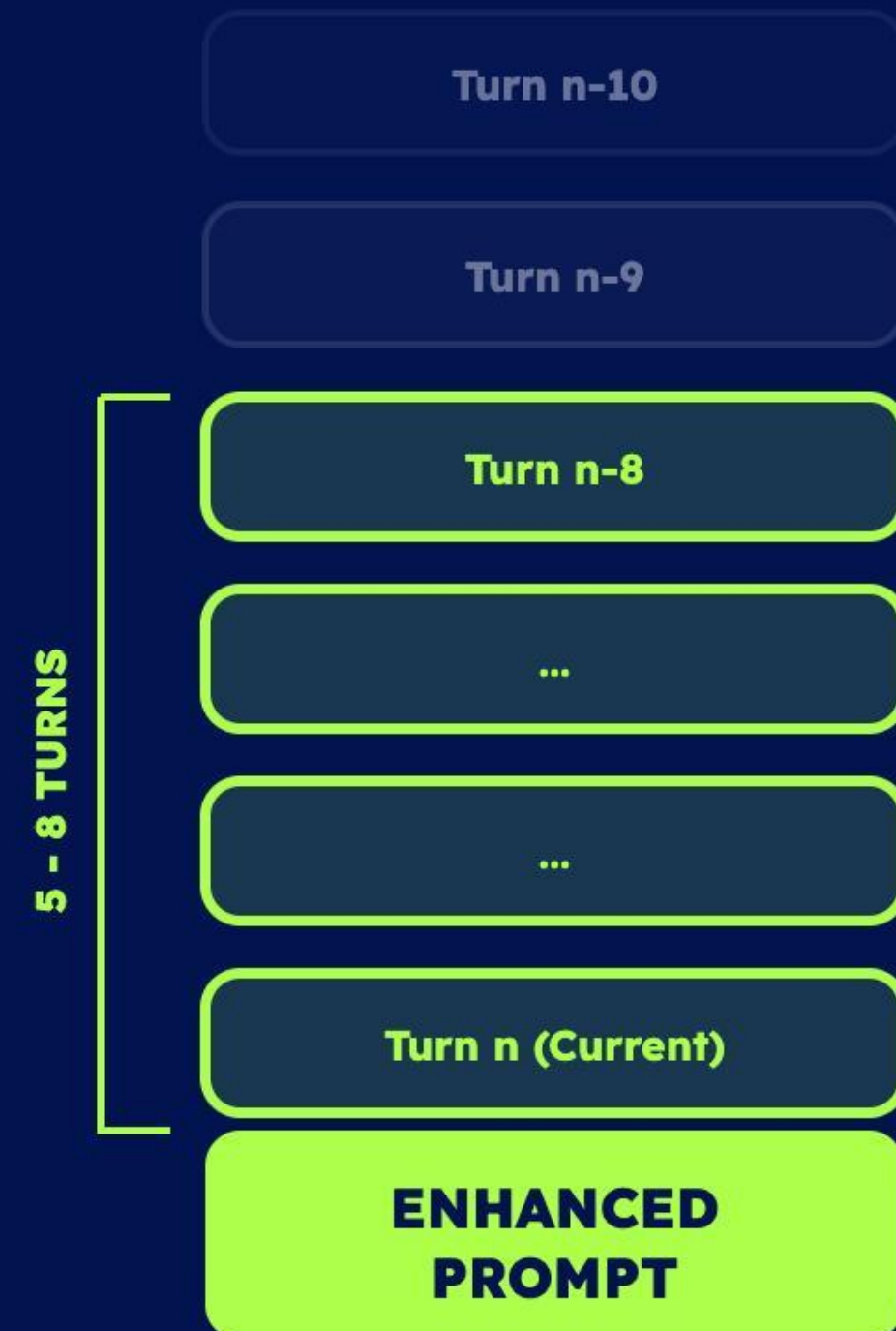
Implement a rolling-window context mechanism:

- Append the last 5–8 conversation turns to each prompt
- Apply minor tweaks to the prompt construction logic

Advantages

- Zero model replacement required
- Zero additional participant recruitment
- Zero changes to the experimental procedure

ROLLING-WINDOW CONTEXT



➔ Improvement 2 - Design Counterbalancing

Solution

Randomize task order across all participants:

- Adopt a Latin Square or full counterbalancing design
- Update task-sequencing logic in the experimental script

Advantages

- Effectively eliminates order bias and halo effects
- Requires no added recruitment burden
- Requires no procedural changes to the tasks themselves

LATIN SQUARE MATRIX

Group A	Task 1	Task 2	Task 3
Group B	Task 3	Task 1	Task 2
Group C	Task 2	Task 3	Task 1

➔ Improvement 3 - Model Upgrades & Validation

Solution

Scale up the LLM architecture via two viable paths:

- Path 1: Local Upgrade

Deploy a larger quantized local model (13B–70B)

Advantage: Strictly preserves data privacy

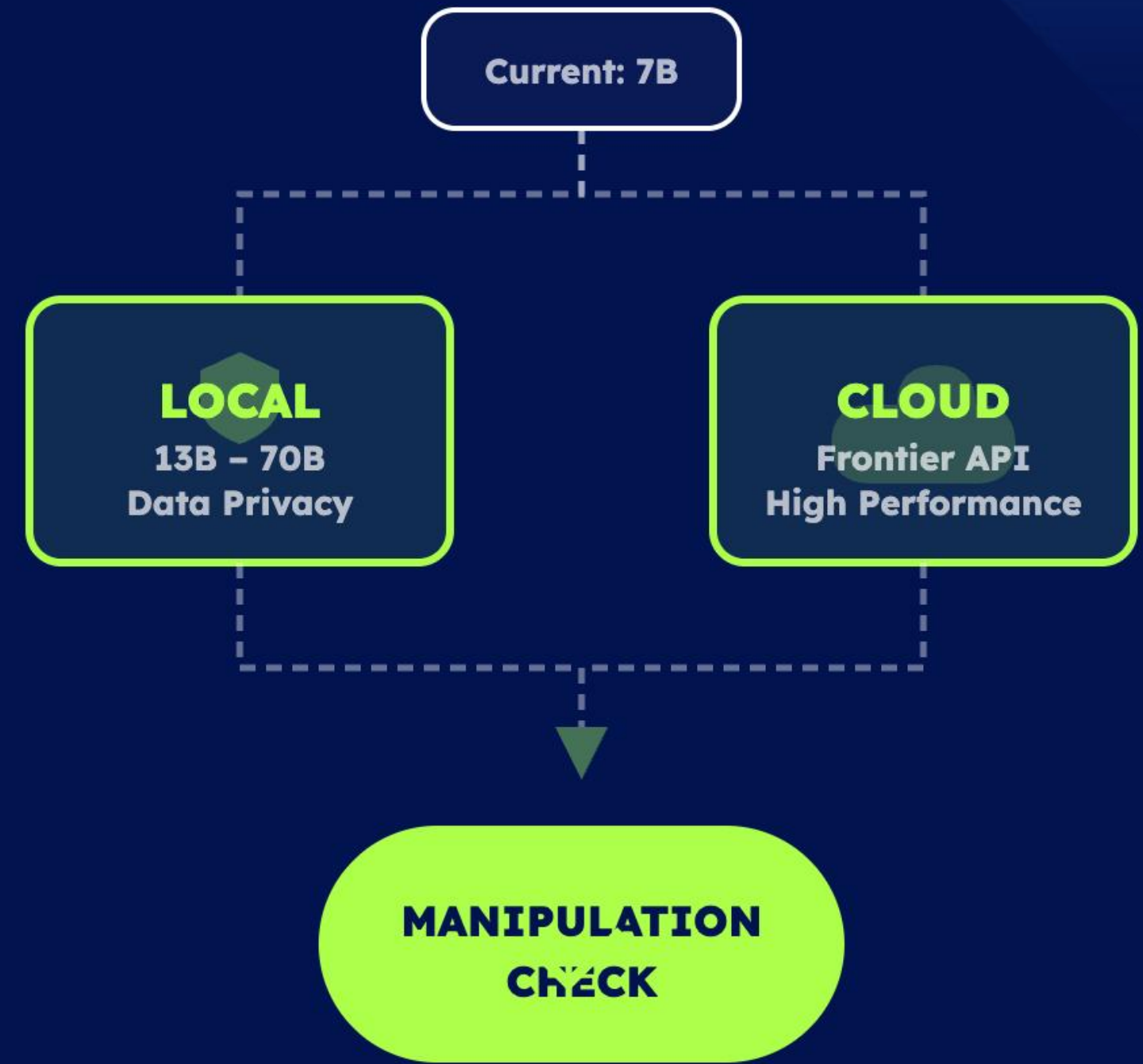
- Path 2: Cloud Transition

Transition to an API-based frontier model

Requirement: Subject to a new ethics review

Mandatory Validation

- Must include a capability manipulation check
- Confirm persona effects are independent of model scale



Q&A

QUESTIONS & ANSWERS

Thank You

Presenters: Dice